

TEST OF MATHEMATICS FOR UNIVERSITY ADMISSION

PAPER 1: APPLICATIONS OF MATHEMATICAL KNOWLEDGE

Practice paper · Set 08

75 minutes

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but do not open the question paper until you are told that you may do so.

A separate answer sheet is provided for this paper. You require a soft pencil and an eraser. Please complete the answer sheet with your candidate number, centre number, date of birth and full name.

This paper contains **20 questions**. For each question, choose the **one** answer you consider correct and record your choice on the separate answer sheet. If you make a mistake, erase thoroughly and try again.

There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all 20 questions. Each question is worth one mark.

You can use the question paper for rough working or notes, but no extra paper is allowed.

You must complete the answer sheet within the time limit.

Calculators and dictionaries are NOT permitted.

There is no formulae booklet for this test.

Please wait to be told you may begin before turning this page.

This is a practice paper generated from a question bank written to the TMUA content specification. It is not an official UAT-UK paper. Answers and worked solutions are in the accompanying solutions booklet.

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1 Simplify $\frac{(2x^3)^4 \times 8x^{-5}}{4x^2}$.

A $32x^9$

B $16x^7$

C x^5

D $16x^5$

E $4x^5$

F $32x^5$

2 Solve $4^{x+1} = 8^{2x-3}$.

A $x = \frac{11}{4}$

B $x = \frac{5}{4}$

C $x = \frac{9}{4}$

D $x = \frac{7}{2}$

E $x = 2$

F $x = 4$

- 3** The quadratic $x^2 + px + q$ has minimum value -4 and one of its roots is 0. Which of the following could be the quadratic?

A $x^2 - 6x + 8$

B $x^2 - 2x - 3$

C $x^2 + 2x - 8$

D $x^2 - 4x$

E $x^2 + 6x + 5$

4 Given that $x^2 + y^2 = 10$ and $x + y = 4$, find the value of xy .

A 4

B 2

C 3

D 6

E -3

F 5

- 5 Two numbers have sum 10 and the sum of their squares is 58. What is the positive difference between them?
- A 3
 - B 4
 - C 6
 - D 1
 - E 2
 - F 8

- 6** The probabilities that a train is early, on time or late are 0.1 , x and $2x$ respectively. Find the probability that the train is not late.
- A** 0.5
 - B** 0.9
 - C** 0.7
 - D** 0.4
 - E** 0.6
 - F** 0.3

- 7 From the top of a cliff 40 m high, the angle of depression of a boat is 30° . How far is the boat from the foot of the cliff?

A 20 m

B $20\sqrt{3}$ m

C $40\sqrt{3}$ m

D 40 m

E 80 m

F $\frac{40}{\sqrt{3}}$ m

- 8** The graph of $y = f(x)$ passes through $(4, 6)$. Which point must lie on the graph of $y = 3f(2x) - 1$?
- A** $(2, 11)$
 - B** $(2, 17)$
 - C** $(8, 17)$
 - D** $(8, 18)$
 - E** $(4, 17)$
 - F** $(2, 18)$

- 9** Solve $2x + 3y = 12$ and $x^2 - y^2 = 0$. How many solutions (x, y) are there?
- A** 0
 - B** 2
 - C** 1
 - D** 4
 - E** 3
 - F** infinitely many

- 10** A trapezium has parallel sides of length 8 cm and 14 cm and area 77 cm^2 . Find the perpendicular distance between the parallel sides.
- A** 14 cm
 - B** 6 cm
 - C** 7 cm
 - D** 11 cm
 - E** 3.5 cm
 - F** 5.5 cm

11 $p(x)$ is a cubic polynomial with $p(1) = p(2) = p(3) = 0$ and $p(0) = 12$. Find $p(4)$.

A 24

B -24

C 12

D 0

E -12

F 6

12 Simplify $\frac{3}{\frac{x+2}{5x-1}} - \frac{2}{x-1}$.

A $\frac{(x+2)(x-1)}{x+1}$

B $\frac{(x+2)(x-1)}{1}$

C $\frac{x+3}{x-7}$

D $\frac{(x+2)(x-1)}{x-1}$

E $\frac{(x+2)(x-1)}{1}$

F $\frac{(x+2)(x-1)}{(x+2)(x-1)}$

13 $PQRS$ is a cyclic quadrilateral with angle $P = 3x$, angle $R = x + 40^\circ$. Find x .

A 45°

B 50°

C 35°

D 30°

E 40°

F 25°

14 How many solutions does $\sin 2x = \frac{1}{2}$ have for $0 \leq x < 2\pi$?

A 2

B 6

C 4

D 8

E 1

F 3

15 The perpendicular bisector of the line segment joining (1, 5) and (7, 1) has equation

A $3x - 2y = 6$

B $2x - 3y = -1$

C $3x + 2y = 18$

D $2x + 3y = 17$

E $y = \frac{3}{2}x - 3$

F $3x - 2y = 18$

16 Solve $\log_3 (x + 6) + \log_3 (x - 2) = 2$.

A $x = 5$

B $x = -7$ or $x = 3$

C $x = 3$

D $x = 3$ or $x = 7$

E $x = -3$

F $x = 1$

17 Given that $5^x = 2$ and $2^y = 125$, find the value of xy .

A 1

B 3

C 5

D 2

E 6

F $\log_5 2 \cdot \log_2 125$

18 The lines $y = 2x + 1$, $y = -x + 7$ and the x -axis enclose a triangle. Find its area.

- A** $\frac{25}{2}$
- B** $\frac{75}{4}$
- C** $\frac{15}{2}$
- D** $\frac{35}{2}$
- E** 15
- F** $\frac{45}{4}$

19 Find the length of the tangent from the point $(7, 1)$ to the circle $x^2 + y^2 - 2x - 4y - 4 = 0$.

A 3

B $\sqrt{28}$

C 6

D $\sqrt{26}$

E $\sqrt{37}$

F 5

20 How many terms in the expansion of $(x + y + z)^4$ (after collecting like terms) are there?

A 16

B 15

C 10

D 5

E 12

F 81

END OF TEST